

market, is ideal for use in solar- and battery-powered LED systems for transport and traffic applications like e-cars, streetlights and traffic signals, as well as marine and air traffic lighting.

Among other solutions, the latest from Texas Instruments (TI) is TPS92210—a natural PFC lighting driver controller with advanced energy features to provide high-efficiency control for LED lighting applications. It offers a predictable maximum power threshold and timed response to an overload, allowing safe handling of surge power requirements.

Early last year, Linear Technology launched LT3791—a synchronous buck-boost DC-DC LED driver and voltage controller that can deliver over 100 watts of LED power. Its 4.7-60V input voltage range makes it ideal for a wide variety of applications including automotive, industrial and architectural lighting. Similarly, its output voltage can be set from 0V to 60V, enabling it to drive a wide range of LEDs in a single string. Its internal four-switch buck-boost controller operates off input voltages above, below or equal to the output voltage—making it ideal for automotive applications, where input voltage can vary dramatically during stopping or starting, cold cranking and load dumping.

Power Integrations' DER-323 reference design for an 18W, 88 per cent efficient, non-isolated A19 LED driver is based on LNK460VG—a low-profile IC from the company's LinkSwitch PL family. The design is suitable for 100W incandescent bulb replacement and is optimised for low line (90V AC to 135V AC) operation. The DER-323 utilises a simple buck-boost converter topology that fits easily inside the A19 bulb. The design has a high power factor of over 0.98, easily satisfying commercial and consumer compliance requirements.

Colour change. LED drivers can also be used to enable colour change or sequencing. This can be achieved

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RBD series LED driver by RECOM Asia

by dimming a mix of coloured LEDs in an array. Another option is that the driver can work with a colour sequencer, which receives the 10V or 24V LED driver output and converts it into a 3-channel output—usually RGB (red, blue and green)—that can be mixed to create a wide, dynamic range of colours.

Resistance to voltage fluctuations. Today, LED drivers are more flexible in handling voltage fluctuations and can even resist these fluctuations.



(L) KC24H-R series constant-current LED driver by Mornsun and (R) GreenChip LED driver SSL21101 by NXP Semiconductors

With GlacialTech's new series of high input-output voltage LED drivers, designers can use as many LEDs in a series as the design requires. Even if the output voltage ranges from 100V to 425V and the input voltage ranges from 90V to 425V, these LED drivers can control and resist any voltage fluctuation. They are very compact in size, which makes LED lighting installation easy.

Flyback and LLC topology. GlacialTech has also introduced an LED driver in flyback topology, which is comparatively compact and cheap. The flyback driver can be used in both AC-DC and DC-DC conversion with galvanic isolation between input and output. These drivers use flyback converter, which is a buck-boost converter with inductor split to form a transformer. This helps to multiply voltage ratios with an additional advantage of isolation.

LED drivers with line-level control (LLC) resonant topology are costlier but give higher efficiency. LLC resonant topology allows zero-voltage switching of main switches, thereby lowering energy loss and boosting efficiency. With LLC resonant converters, an efficiency of 93-96 per cent can be achieved.

TPS92020 from TI is a high-performance, resonant-switching LED driver controller. It is designed for use in high-power LED lighting systems. It uses resonant switching in LLC topology.

